# Qiskit Fall Fest 2026

## Challenge 2: Make the Quantum Coin Biased

**Problem Statement :** A Hadamard gate creates an equal superposition of |0⟩ and |1⟩, producing a result similar to a fair quantum coin. But can you control the measurement probabilities? Your task is to modify a single-qubit circuit so that the measurement outcomes are not equally distributed.

**Your Task -** Using Qiskit:

1. Create a single-qubit quantum circuit.
2. Start with the qubit in |0⟩.
3. Apply quantum gates to create a state whose measurement probabilities for 0 and 1 are different.
4. Measure the qubit.
5. Execute the circuit for 100 shots.
6. Observe the number of 0 and 1 outcomes.
7. Display the results using a histogram.
8. Try to create different degrees of bias.

**Think Before You Run -** Before executing your circuit, predict:

- Will 0 or 1 occur more frequently?
- Approximately what percentage of the measurements will produce each outcome?
- What happens if you change the gates in your circuit?

**Design Constraints -** Your circuit must:

- Use exactly 1 qubit.
- Include measurement.
- Use no more than 4 quantum gates before measurement.
- Be created and executed using Qiskit.
- Use 100 shots.

**Bonus Challenge** ⭐ **-** Try to design two different circuits:

- One where 0 occurs more frequently.
- One where 1 occurs more frequently.

Compare their histograms.

**Submission -** Your notebook must include:

- Your quantum circuit.
- Your predicted probabilities.
- Measurement results for 100 shots.
- Histogram of the results.
- A short explanation of how your gates changed the measurement distribution.

**Challenge Question – (place the answer in the last cell of your notebook)**

How can quantum gates change the probability of obtaining 0 or 1 when a qubit is measured?